# Psychotropic Medication

**Psychiatry, Clinical Lecture** | Dr. Hussain M. Khrad (Consultant Psychiatrist, KFSHRC-Jeddah) | Source: `Psychotropic_medication_2.pdf` (73 slides)

> **Legend:** ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture. All MCQ answers and both case model answers come from the lecture's own answer slides.
>
> 🔗 **Bridge from the disorder lectures (this is the drug lecture they all point to):**
>
> - **Mood:** SSRIs first-line; **4–6 weeks**; continue 6–9 months; **lithium = anti-suicide**; avoid antidepressant monotherapy in bipolar; **TCAs lethal in overdose**.
> - **Anxiety/OCD:** SSRI/SNRI first-line; **high-dose SSRI for OCD**; clomipramine; **buspirone** adjunct; **benzodiazepines short-term**; propranolol for performance anxiety; **prazosin** for PTSD nightmares.
> - **Psychosis:** the **four dopamine pathways**, FGA vs SGA, the **EPS clock**, **NMS**, **clozapine** (ANC, myocarditis, ileus, 48-hour rule), prolactin, QT.
> - **Substance Use:** benzodiazepine withdrawal **kills**; **flumazenil → seizures**; bupropion and varenicline for smoking.
> - **Medicine:**
> - **ECG** (QT/QRS; TCA overdose → tall R in aVR)
> - **Pituitary I** (prolactin)
> - **Thyroid/Parathyroid** (lithium → hypothyroidism, ↑ Ca/PTH)
> - **Posterior Pituitary** (lithium → **nephrogenic DI**; SSRIs → **SIADH**)

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## 1. Title / Objectives

| Domain | Objective | Likely tested as |
|---|---|---|
| Knowledge | ⭐ **Four workhorse classes** and how they act | Antidepressants · anxiolytics/sedatives · antipsychotics · mood stabilisers |
| Knowledge | Common and serious side-effects by class | SIADH, QT, sexual dysfunction, prolactin, metabolic |
| Knowledge | ⭐ **Serotonin syndrome, NMS, lithium toxicity** | Tramadol + SSRI; haloperidol + rigidity; NSAID + lithium |
| Knowledge | EPS timeline; lithium range, monitoring and interactions; **black-box warnings** | — |
| Skill | Choose an antidepressant by **side-effect profile and overdose safety** | No TCAs for the suicidal |
| Skill | ⭐ **Serotonin syndrome vs NMS** at the bedside | Clonus + fast vs rigidity + slow |
| Skill | ⭐ **Reconcile psychotropics on admission; never stop abruptly** | Diazepam omitted → seizure |
| Skill | Spot everyday interactions; baseline + monitoring; ⭐ **fever/sore throat on clozapine → ANC the same day** | — |

| Domain | Objective | Likely tested as |
|---|---|---|
| Values | Every prescription is an efficacy–harm trade-off; ask about **sexual and metabolic** effects; consent = what it does **and costs**; never give a lethal drug to a patient at risk; involve the patient | — |

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# 2. Main Content (Lecture Order)

# PART 1: ANTIDEPRESSANTS

## 2.1 Classes and Principles (Slides 5–6)

**Classes:**

- **SSRIs**
- **SNRIs**
- **heterocyclics** (TCAs, tetracyclics)
- **MAOIs**
- **miscellaneous** (bupropion, mirtazapine, trazodone, newer agents)

**Principles:**

- ⭐ **similar efficacy across classes**: they differ in **safety and side-effects**
- ⭐ **~60–70% respond**; full benefit at **4–6 weeks** on a given dose
- **start low, titrate slowly** in the side-effect-prone

| Choosing (slide 6) | Factors |
|---|---|
| **Patient & illness** | Target symptoms (insomnia, low appetite); ⭐ **prior response in the patient or a family member**; comorbidities |
| **Drug** | Side-effect profile; interactions; ⭐ **overdose risk (TCAs are lethal)** |
| **Practical** | Cost; adherence and dosing convenience |

⭐ **A suicidal patient should not be given a drug that is lethal in overdose.**

## 2.2 SSRIs (Slides 7–8) ⭐⭐

**First-line because they work and are safe in overdose.**

| Agent | Signature |
|---|---|
| ⭐ **Fluoxetine** | **Long half-life: no taper needed** (*[added]* also the longest MAOI washout, 5 weeks) |
| ⭐ **Sertraline** | **Fewest interactions; safe in cardiac disease** |
| ⭐ **Citalopram / escitalopram** | **Most QT-prolonging** |
| ⭐ **Paroxetine** | **Short half-life: worst discontinuation** |
| ⭐ **Fluvoxamine** | **CYP1A2 inhibitor → raises clozapine** |

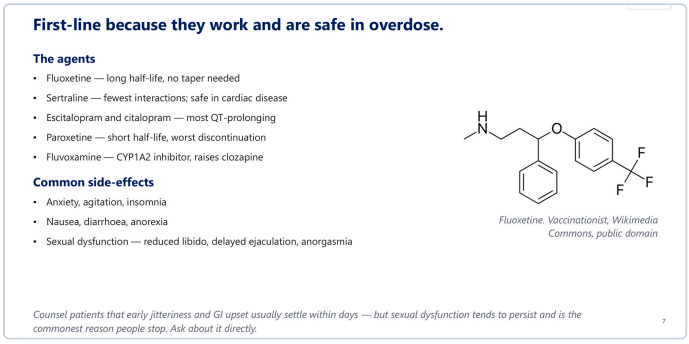
**Common side-effects:**

- anxiety, agitation, insomnia
- nausea, diarrhoea, anorexia
- ⭐ **sexual dysfunction** (↓ libido, delayed ejaculation, anorgasmia)

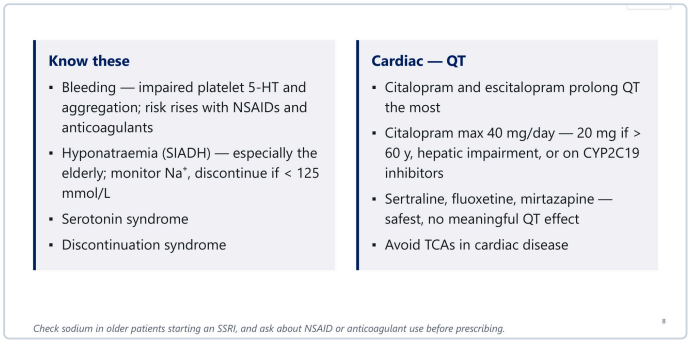
Early jitteriness and GI upset settle within days; **sexual dysfunction persists and is the commonest reason people stop**. Ask directly.

| Serious effects | Cardiac (QT) |
|---|---|
| ⭐ **Bleeding**: impaired platelet 5-HT; ↑ with **NSAIDs, anticoagulants** | ⭐ **Citalopram/escitalopram prolong QT the most** |
| ⭐ **Hyponatraemia (SIADH)**: elderly; monitor $Na^+$; **stop if < 125 mmol/L** | ⭐ **Citalopram max 40 mg/day; 20 mg if > 60 y**, hepatic impairment, or on CYP2C19 inhibitors |
| **Serotonin syndrome** | ⭐ **Sertraline, fluoxetine, mirtazapine: safest**, no meaningful QT effect |
| **Discontinuation syndrome** | **Avoid TCAs in cardiac disease** |

**Check sodium in older patients; ask about NSAIDs/anticoagulants.**




*Slide 7: SSRIs*




*Slide 8: SSRI serious effects*

## 2.3 SNRIs (Slide 9)

| Agent | Flags |
|---|---|
| **Venlafaxine / desvenlafaxine** | ⭐ **Dose-dependent hypertension**: check BP. ⭐ **One of the worst discontinuation syndromes** (short half-life) |
| **Duloxetine** | Also for **neuropathic pain and fibromyalgia**; **avoid in severe liver disease** |

**Dual 5-HT/NA action without the TCA receptor baggage.**

## 2.4 Tricyclic Antidepressants (Slides 10–12) ⭐⭐

| Receptor blocked | Effects | Notes |
|---|---|---|
| **Antihistaminic (H1)** | Sedation, weight gain | Often why patients stop |
| **Anticholinergic (M)** | Dry mouth, constipation, **urinary retention, blurred vision** | Worse in the elderly; **avoid in prostatism and glaucoma** |
| **Anti-adrenergic (α1)** | **Orthostatic hypotension** | **Falls** risk |
| ⭐ **Cardiac ($Na^+$ channel)** | ⭐ **QRS widening, QT prolongation, arrhythmia** | ⭐ **LETHAL IN OVERDOSE**: never prescribe freely to a suicidal patient |

| Tertiary amines (more side-effects) | Secondary amines (milder) |
|---|---|
| **Imipramine, amitriptyline, doxepin, clomipramine** | **Desipramine, nortriptyline** (metabolites of the tertiaries) |

**Mnemonic *[added]* for TCA toxicity, "3 C's": C**onvulsions, **C**oma, **C**ardiotoxicity (+ anticholinergic toxidrome).

**Case (slides 11–12):**

- 22-year-old, **2 h after a bottle of amitriptyline**
- drowsy, **BP 88/54, HR 128**
- ⭐ **QRS 130 ms, tall R in aVR**

→ ⭐ **B: IV sodium bicarbonate.**

- TCA toxicity = **$Na^+$-channel blockade**; **QRS > 100 ms and a tall R in aVR predict seizures and arrhythmia**.
- **Bicarbonate overcomes it.**
- Charcoal alone doesn't treat the cardiotoxicity; ⭐ **flumazenil is wrong and dangerous** (*[added]* it unmasks seizures).

| Receptor blocked | Effects | Notes |
|---|---|---|
| **Antihistaminic** | Sedation, weight gain | Often the reason patients stop |
| **Anticholinergic** | Dry mouth, constipation, urinary retention, blurred vision | Worse in the elderly; avoid in prostatism and glaucoma |
| **Antiadrenergic ($\alpha_1$)** | Orthostatic hypotension | Falls risk |
| **Cardiac ($Na^+$ channel)** | QRS widening, QT prolongation, arrhythmia | LETHAL IN OVERDOSE — never prescribe freely to a suicidal patient |
| **Tertiary vs secondary** | Tertiary: imipramine, amitriptyline, doxepin, clomipramine — more side-effects | Secondary: desipramine, nortriptyline — metabolites of the tertiary amines, generally milder |

*Effective but toxic. Reserve TCAs, and remember they are the overdose-lethal class.*

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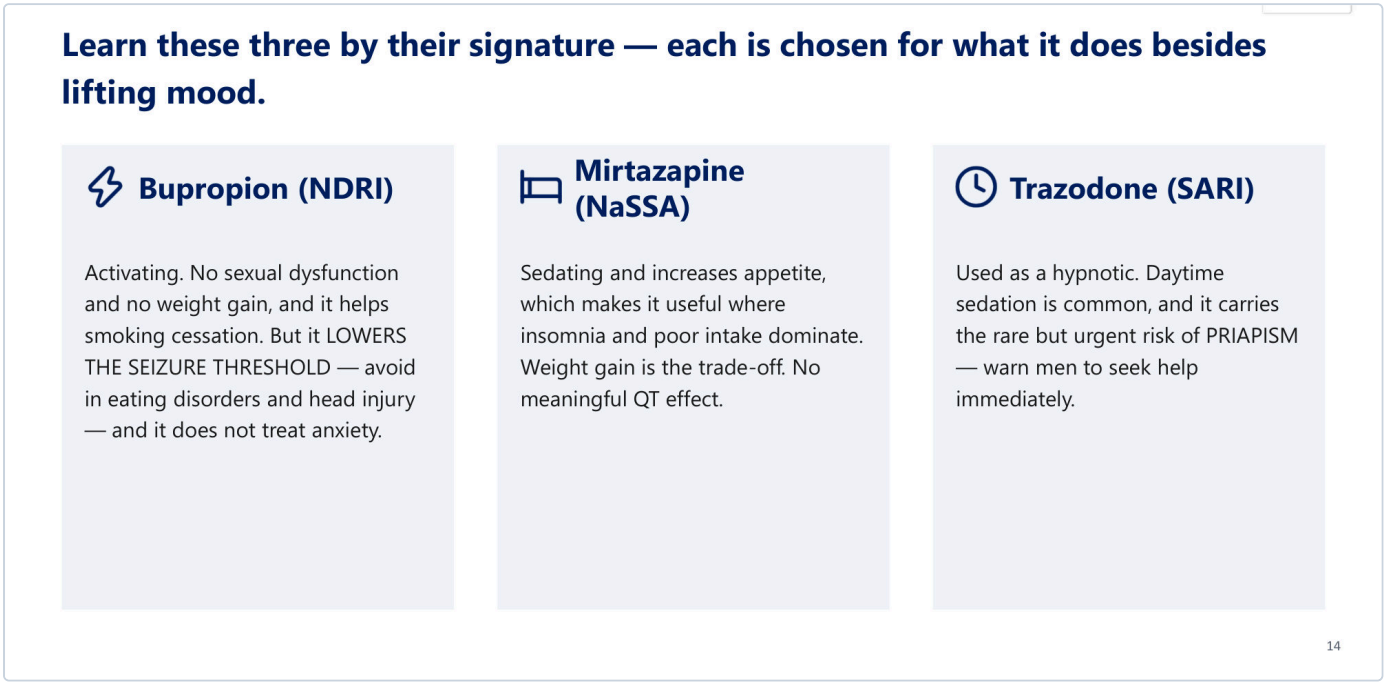
*Slide 10: TCA receptor effects*

## 2.5 MAOIs (Slide 13)

- **Background:**
  - **iproniazid** (an isoniazid derivative) lifted mood in the 1950s
  - agents: **isocarboxazid, tranylcypromine, phenelzine**; ⭐ **moclobemide is reversible (RIMA)**
  - very effective, but reserved for **atypical or treatment-resistant** depression
- **The two big dangers:**
  1. ⭐ **Hypertensive crisis with tyramine** (aged cheese, cured meats; the **"cheese reaction"**) or sympathomimetics
  2. ⭐ **Serotonin syndrome** with serotonergic drugs
- ⭐ **Washout: 2 weeks, or 5 weeks after fluoxetine.**
- **Also:** orthostatic hypotension, weight gain, sexual dysfunction, insomnia.

## 2.6 Bupropion, Mirtazapine, Trazodone (Slide 14) ⭐

| Drug | Class | Signature |
|---|---|---|
| ⭐ **Bupropion** | **NDRI** | **Activating**; ⭐ **no sexual dysfunction, no weight gain**; **smoking cessation**. ⭐ **LOWERS THE SEIZURE THRESHOLD**: avoid in **eating disorders and head injury**. **Doesn't treat anxiety** |
| ⭐ **Mirtazapine** | **NaSSA** | **Sedating, ↑ appetite**: useful when **insomnia and poor intake** dominate. Weight gain. **No QT effect** |
| ⭐ **Trazodone** | **SARI** | Used as a **hypnotic**; daytime sedation. ⭐ **PRIAPISM**: warn men to seek help immediately |




*Slide 14: Bupropion, mirtazapine, trazodone*

## 2.7 Newer Antidepressants (Slide 15)

| Serotonin-modulating | Rapid-acting / novel |
|---|---|
| **Vortioxetine**: multimodal, **pro-cognitive** | ⭐ **Esketamine (intranasal)**: **NMDA antagonist**; treatment-resistant depression and **acute suicidality**; now also **monotherapy (2025)** |
| **Vilazodone**: SPARI | **Dextromethorphan–bupropion (Auvelity, 2022)**: rapid oral |
| **Agomelatine**: **MT1/MT2 agonist + 5-HT2C antagonist** | ⭐ **Zuranolone (2023)**: oral, **postpartum depression** |

The rapid-acting group works **far faster than 4–6 weeks**.

## 2.8 Serotonin Syndrome (Slides 16–19) ⭐⭐

**Case (slides 16–17):**

- 30-year-old on **sertraline** given **tramadol**
- within **hours**: agitated, confused, sweating, **39 °C**, tachycardia, **brisk reflexes and clonus worse in the legs**

→ ⭐ **B: serotonin syndrome.**

**Triad:**

| Mental status | Neuromuscular | Autonomic |
|---|---|---|
| Agitation, confusion | ⭐ **Hyperreflexia, clonus (esp. lower limbs)**, myoclonus, **ocular clonus**, shivering, tremor | Tachycardia, **hyperthermia**, ↑ **GI motility** (*[added]* diarrhoea, mydriasis) |

- **Culprits:**
  - SSRIs/SNRIs **with tramadol, meperidine (pethidine), fentanyl**
  - **triptans**
  - ⭐ **linezolid**
  - **MAOIs**
  - **ondansetron, metoclopramide**
  - **dextromethorphan**
  - **St John's wort**
- **Management:**
  1. **stop the offending agents**
  2. supportive care, **cooling**
  3. **benzodiazepines**
  4. ⭐ **cyproheptadine (5-HT2A antagonist)** if severe
- ⭐ **Hidden serotonergic drugs: tramadol, dextromethorphan, linezolid, triptans, St John's wort**.

**Serotonin Syndrome vs NMS (Slide 19) ⭐⭐**

| Feature | Serotonin syndrome | NMS |
|---|---|---|
| Cause | **Serotonergic** drugs | **Dopamine antagonists** (or **dopaminergic withdrawal**, e.g. stopping L-dopa) |
| Onset | ⭐ **Fast: within 24 h** | ⭐ **Slow: days to weeks** |
| Neuromuscular | ⭐ **Hyperreflexia, clonus, myoclonus** | ⭐ **"Lead-pipe" rigidity, bradyreflexia** |
| Shared | **Hyperthermia · autonomic instability · altered mental state · ↑ CK** | Same |
| Treatment | **Benzodiazepines + cyproheptadine** | **Stop drug + dantrolene/bromocriptine + benzodiazepines** |

⭐ **Reflexes UP + FAST = serotonin syndrome; RIGIDITY + SLOW = NMS.** From the case slide: **dilated pupils + hyperactive bowel** (SS) vs **quiet bowel** (NMS).

| Feature | Serotonin syndrome | Neuroleptic malignant syndrome |
|---|---|---|
| **Cause** | Serotonergic drugs | Dopamine antagonists (or dopaminergic withdrawal) |
| **Onset** | Fast — within 24 hours | Slow — days to weeks |
| **Neuromuscular** | Hyperreflexia, clonus, myoclonus | "Lead-pipe" rigidity, bradyreflexia |
| **Shared** | Hyperthermia · autonomic instability · altered mental state · raised CK | Same |
| **Treatment** | Benzodiazepines + cyproheptadine | Stop drug + dantrolene / bromocriptine + benzodiazepines |

*Quick discriminator: reflexes up and fast onset = serotonin syndrome; rigidity and slow onset = NMS.*

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*Slide 19: Serotonin syndrome vs NMS*

## 2.9 Discontinuation Syndrome (Slides 20–22) ⭐

Not relapse, not addiction: a **self-limiting withdrawal**.

- **worst with short half-life agents (paroxetine, venlafaxine)**
- **onset within days; resolves in < 2 weeks**
- **taper** to prevent it

**FINISH:**

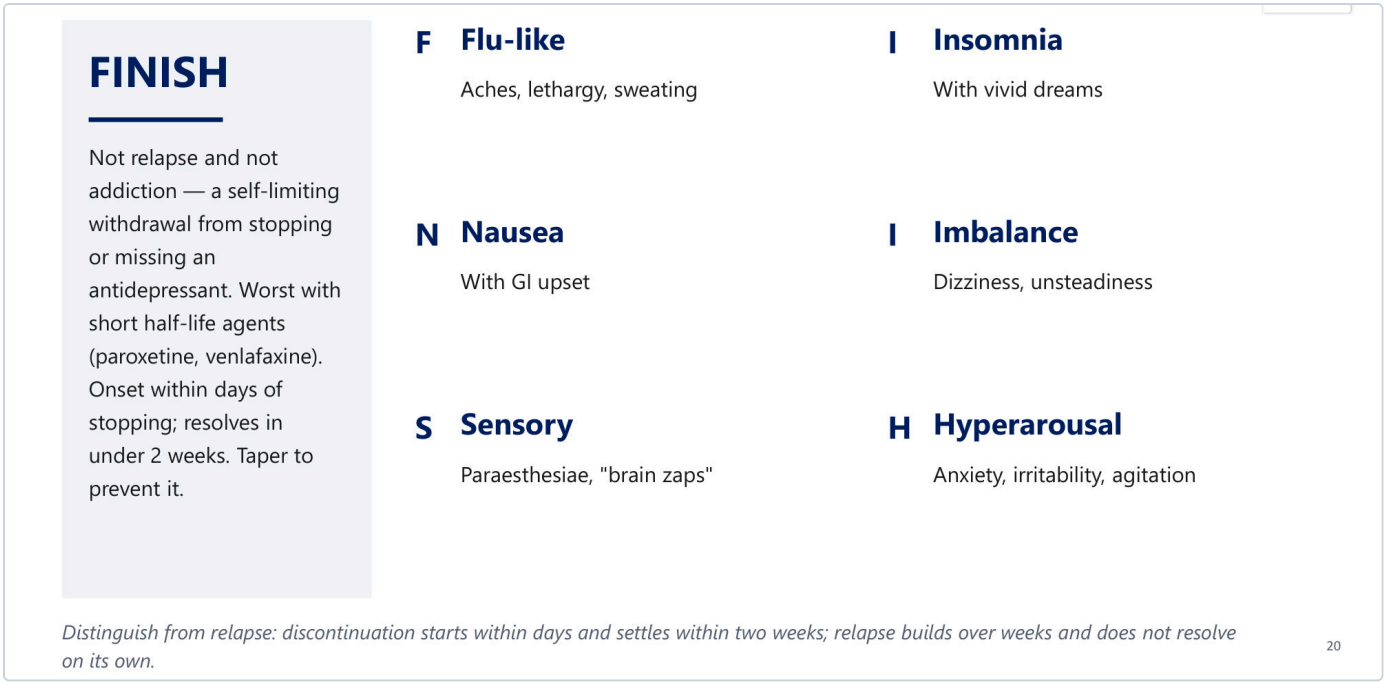
- **F**lu-like (aches, lethargy, sweating)
- **I**nsomnia (vivid dreams)
- **N**ausea/GI upset
- **I**mbalance (dizziness)
- **S**ensory (paraesthesiae, ⭐ **"brain zaps"**)
- **H**yperarousal (anxiety, irritability)

**Case (slides 21–22):**

- well for 8 months; **stopped paroxetine abruptly**
- **3 days later:** dizziness, nausea, **electric shocks in the head**, vivid dreams, irritability
- **mood unchanged**

→ ⭐ **B: discontinuation syndrome. Restart, then taper properly.**

| | Discontinuation | Relapse |
|---|---|---|
| Onset | **Days** after stopping | **Weeks** |
| Leads with | **Physical** symptoms | **Low mood, anhedonia** |
| Course | Settles in **< 2 weeks** | Doesn't resolve on its own |



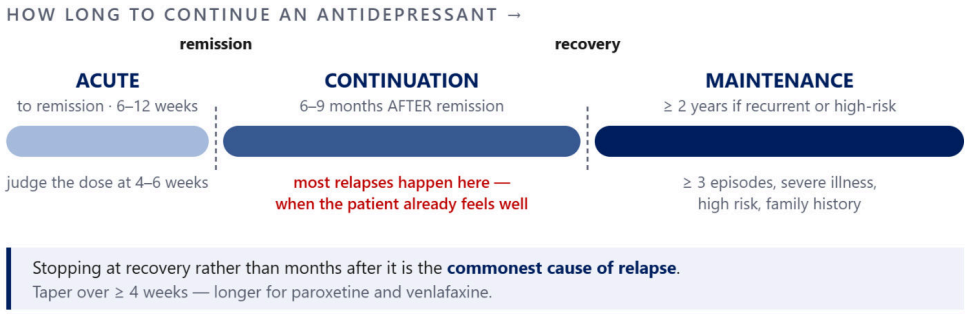

*Slide 20: FINISH*

## 2.10 Duration and Pregnancy (Slides 23–24) ⭐

| Phase | Duration | Note |
|---|---|---|
| **Acute** | To remission: **6–12 weeks** | Judge the dose at **4–6 weeks** |
| ⭐ **Continuation** | **6–9 months AFTER remission** | ⭐ **Most relapses happen here**, when the patient already feels well |
| **Maintenance** | ⭐ **≥ 2 years** if recurrent/high-risk | **≥ 3 episodes**, severe illness, high risk, family history |

- ⭐ **Stopping at recovery rather than months after it is the commonest cause of relapse.**
- **Taper over ≥ 4 weeks**; longer for paroxetine and venlafaxine.
- **Other clocks:**
  - ⭐ **benzodiazepines 2–4 weeks**, not months
  - **antipsychotics 1–2 years** after a first psychotic episode
  - **lithium long-term, never stopped abruptly** (rebound mania)

## Choosing the drug is half the decision — the other half is how long to stay on it.




*The other classes run to different clocks: benzodiazepines 2–4 weeks, not months; antipsychotics 1–2 years after a first episode of psychosis; lithium long-term, and never stopped abruptly — abrupt withdrawal can precipitate rebound mania.*

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*Slide 23: How long to continue treatment*

| Pregnancy | Lactation |
|---|---|
| **SSRIs are not major teratogens**; no robust cardiac link after adjustment | Overall infant risk **low** |
| **Paroxetine:** historic weak cardiac signal; generally **avoided in the first trimester** | ⭐ **Lowest infant levels: sertraline, paroxetine** |
| Late use → transient **neonatal adaptation** syndrome | Higher: **citalopram, fluoxetine, venlafaxine** |
| ⭐ **Untreated severe depression is itself a risk**: blanket stopping is often wrong | — |

🔁 **Recap: Antidepressants.**

- Similar efficacy; 60–70% respond; 4–6 weeks.
- **SSRIs:** fluoxetine (long t½), sertraline (cardiac-safe), citalopram (QT, max 40/20 mg), paroxetine (worst discontinuation), fluvoxamine (↑ clozapine). Risks: sexual dysfunction, bleeding, SIADH.
- **SNRIs:** venlafaxine (BP, discontinuation); duloxetine (pain).
- **TCAs:** H1, M, α1, $Na^+$; lethal → **bicarbonate**.
- **MAOIs:** tyramine; 2/5-week washout.
- **Bupropion** (seizures; no sexual dysfunction); **mirtazapine** (sleep, appetite); **trazodone** (priapism).
- **Serotonin syndrome** (fast, clonus) vs **NMS** (slow, rigid).
- **FINISH** discontinuation.
- Continue 6–9 months; ≥ 2 years if recurrent.

▶️ **Up next: Anxiolytics and sedatives.**

# PART 2: ANXIOLYTICS & SEDATIVES

## 2.11 Benzodiazepines (Slides 26–30) ⭐⭐

**Mechanism:**

- bind **GABA-A**; ⭐ **increase the FREQUENCY** of chloride-channel opening
- ⭐ **barbiturates increase the DURATION**, which is **why they are lethal alone**
- **"Benzos = Frequency; Barbiturates = Duration"**

**Uses:**

- anxiety; **short-term insomnia**
- muscle relaxation
- ⭐ **status epilepticus**
- **rapid tranquillisation**
- ⭐ **alcohol and sedative withdrawal**

| Group | Agents | Why it matters |
|---|---|---|
| **Short-acting** | Alprazolam, lorazepam, oxazepam, temazepam | More **rebound and withdrawal** between doses |
| **Long-acting** | **Diazepam, chlordiazepoxide, clonazepam** | Smoother, but **accumulate**: caution in the elderly |
| ⭐ **"LOT": liver-safe** | **Lorazepam · Oxazepam · Temazepam** | **Phase-II glucuronidation only**: preferred in **liver disease and the elderly** |
| **Reversal** | **Flumazenil** | ⭐ **Can precipitate SEIZURES** in dependence or mixed overdose: rarely used; supportive airway care is the mainstay |

| Side-effects & misuse | ⭐ Withdrawal can be dangerous |
|---|---|
| Sedation, cognitive/psychomotor impairment, **falls in the elderly** | ↑ anxiety, sweats, tremor, hypersensitivity, aches, headache |
| ⭐ **Anterograde amnesia**; **paradoxical disinhibition** | ⭐ **SEIZURES, COMA, DEATH**, like alcohol withdrawal |
| **Tolerance within ~2 weeks** | ⭐ **Never stop abruptly** in a dependent patient: **taper** |

**Pregnancy:**

- avoid if possible
- **cleft palate** signal
- ⭐ **floppy-infant syndrome** and neonatal withdrawal near term
- present in breast milk

**Case (slides 29–30):**

- 58-year-old, elective hernia repair
- **diazepam 20 mg daily for years, not written up**
- **day 3:** tremulous, sweating, **BP 175/100**, then a **GTC seizure**; no alcohol

→ ⭐ **B: benzodiazepine withdrawal → reinstate, then taper. Reconcile psychotropics on every admission.**

| Group | Agents | Behaviour / why it matters |
|---|---|---|
| **Short-acting** | Alprazolam, lorazepam, oxazepam, temazepam | More rebound and withdrawal between doses |
| **Long-acting** | Diazepam, chlordiazepoxide, clonazepam | Smoother cover, but they accumulate — caution in the elderly |
| **"LOT" — liver-safe** | Lorazepam · Oxazepam · Temazepam | Phase-II glucuronidation only, no oxidative metabolism — preferred in liver disease and the elderly |
| **Reversal** | Flumazenil | Reverses overdose but CAN PRECIPITATE SEIZURES in dependence or mixed overdose — used rarely; supportive airway care is the mainstay |

*Half-life predicts behaviour; the LOT drugs are the answer whenever the liver is compromised.*

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*Slide 27: Choosing and reversing benzodiazepines*

## 2.12 Z-Drugs and Buspirone (Slide 31)

| Z-drugs (hypnotics) | Buspirone (anxiolytic) |
|---|---|
| **Zolpidem, zopiclone, zaleplon** | ⭐ **5-HT1A partial agonist** |
| GABA-A benzodiazepine site, more selectively | ⭐ **No dependence, no sedation, no withdrawal** |
| Short-term insomnia, but still **dependence, complex sleep behaviours, falls** | ⭐ **Delayed onset 1–2 weeks: useless for acute anxiety**; an adjunct in **GAD** |

🔁 **Recap: Anxiolytics.** Benzodiazepines = frequency; barbiturates = duration. LOT for liver disease/elderly. Tolerance in ~2 weeks; never stop abruptly (seizures); flumazenil → seizures; 2–4 weeks max. Z-drugs still cause dependence. Buspirone: 5-HT1A, no dependence, slow.

▶️ **Up next: Antipsychotics.**

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# PART 3: ANTIPSYCHOTICS

## 2.13 NMS (Slides 34–35, 44) ⭐⭐

**Case:**

- 24-year-old, **haloperidol 5 days ago**
- **high fever, "lead-pipe" rigidity, fluctuating consciousness, labile BP, tachycardia, markedly ↑ CK**

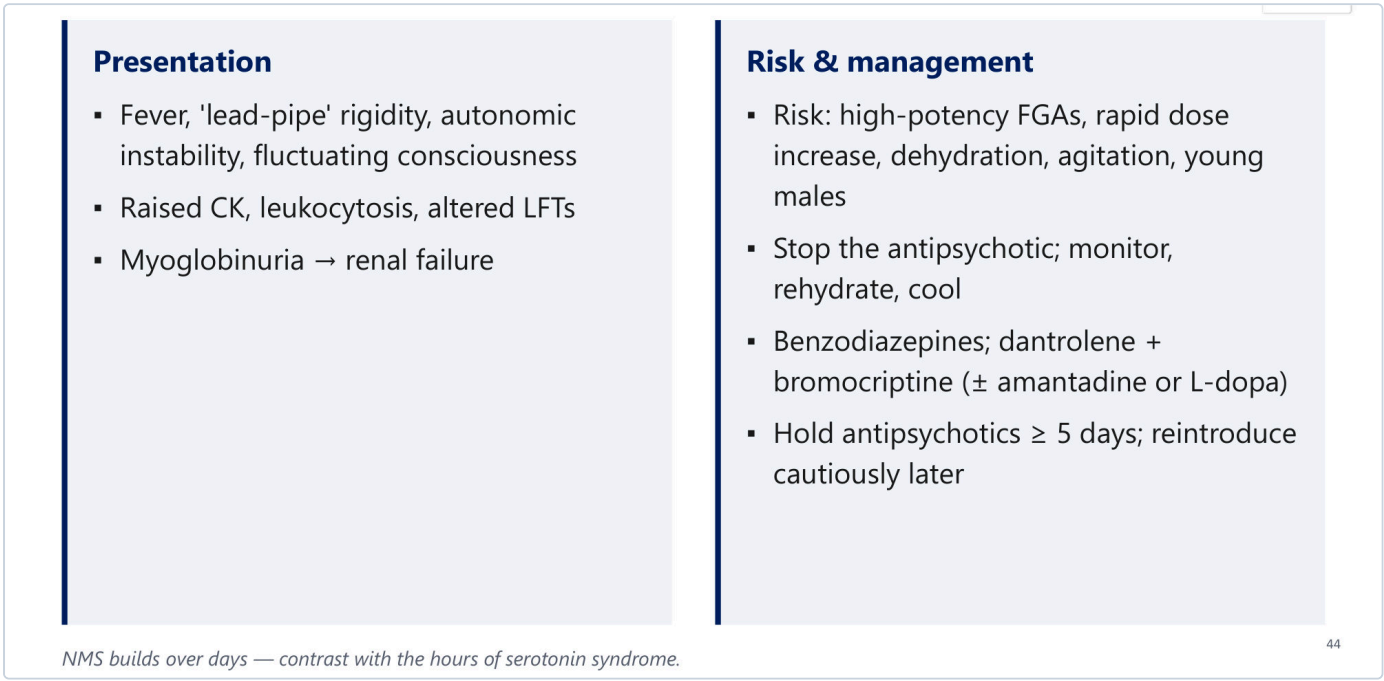
→ ⭐ **B: NMS.**

| Presentation | Risk & management |
|---|---|
| **Fever, lead-pipe rigidity, autonomic instability, fluctuating consciousness** | **Risk: high-potency FGAs, rapid dose increase, dehydration, agitation, young males** |
| ⭐ **↑ CK, leukocytosis, altered LFTs** | ⭐ **Stop the antipsychotic**; monitor, **rehydrate, cool** |

| Presentation | Risk & management |
|---|---|
| ⭐ **Myoglobinuria → renal failure** | **Benzodiazepines; dantrolene + bromocriptine** (± amantadine or L-dopa) |
| — | ⭐ **Hold antipsychotics ≥ 5 days**; reintroduce cautiously |

**Mnemonic *[added]*: "FEVER"**:

- **F**ever
- **E**ncephalopathy
- **V**itals unstable
- **E**levated CK/WCC
- **R**igidity




*Slide 44: NMS*

## 2.14 Mechanism, FGA vs SGA (Slides 36–38)

| Pathway | D2 blockade effect |
|---|---|
| **Mesolimbic** | ⭐ **Therapeutic**: ↓ positive symptoms |
| **Mesocortical** | May **WORSEN** negative/cognitive symptoms |
| **Nigrostriatal** | ⭐ **EPS** |
| **Tuberoinfundibular** | ⭐ **Hyperprolactinaemia** |

| Typical (FGA) | Atypical (SGA) | D2 partial agonists ("third generation") |
|---|---|---|
| **Chlorpromazine, fluphenazine, haloperidol, perphenazine** | **Risperidone, paliperidone, olanzapine, quetiapine, ziprasidone, lurasidone, asenapine, clozapine** | ⭐ **Aripiprazole, brexpiprazole, cariprazine** |
| Mainly **D2 antagonists**; positive symptoms; little effect on negative | **5-HT2A + D2 antagonism** → fewer EPS, some negative-symptom effect, **more metabolic** | A different mechanism; EPS mainly **akathisia** |

| Typical (FGA) | Atypical (SGA) | D2 partial agonists ("third generation") |
|---|---|---|
| ⭐ **High-potency (haloperidol, fluphenazine) → more EPS and prolactin** | ⭐ **No clear efficacy advantage except clozapine** | ⭐ **Lurasidone and cariprazine** also licensed for **bipolar depression** |
| ⭐ **Low-potency (chlorpromazine) → more anticholinergic, sedation, hypotension** | Judge efficacy only after **≥ 4 weeks** at an adequate dose | — |

- **Class risks:** EPS, tardive dyskinesia, NMS.
- **5-HT2A antagonism lifts the striatal brake**: fewer movement disorders, at the cost of weight and metabolic risk.

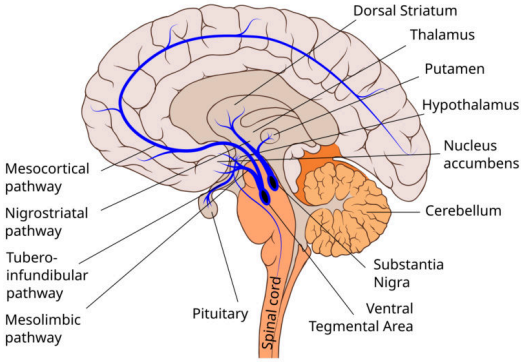
### One diagram explains the whole antipsychotic side-effect story.

#### What D2 blockade does in each

- Mesolimbic → the therapeutic target, reduces positive symptoms
- Mesocortical → blockade may WORSEN negative and cognitive symptoms
- Nigrostriatal → extrapyramidal side-effects
- Tuberoinfundibular → hyperprolactinaemia

#### Typical vs atypical

- Typical (FGA): mainly D2 antagonists
- Atypical (SGA): 5-HT2A + D2 antagonism → fewer EPS, some effect on negative symptoms, more metabolic effects




*Dopaminergic pathways. Slashme / Patrick J. Lynch, Wikimedia Commons, CC BY-SA 4.0*

*Atypicals add 5-HT2A antagonism, which lifts the striatal brake — fewer movement disorders, at the cost of weight and metabolic risk.*

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*Slide 36: Four dopamine pathways*

## 2.15 The Side-Effect Map (Slide 39) ⭐⭐

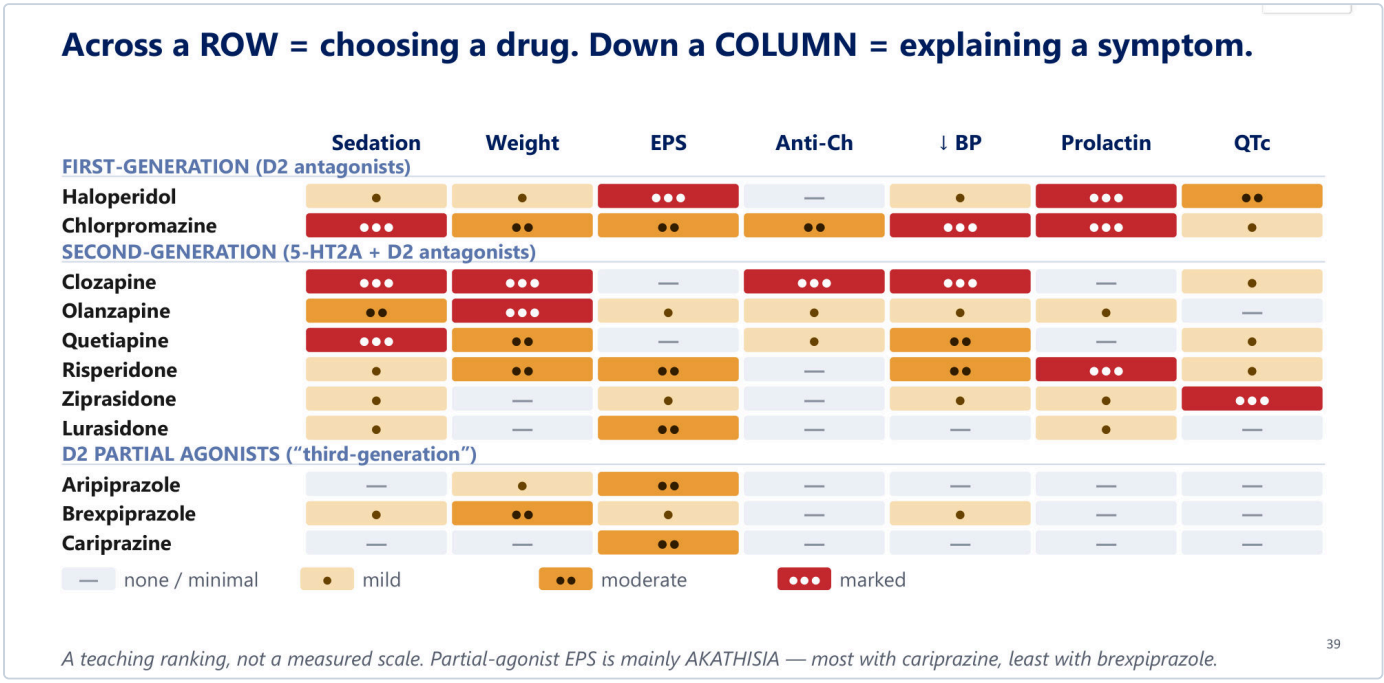
**Across a ROW = choosing a drug; down a COLUMN = explaining a symptom.** (— none, ● mild, ●● moderate, ●●● marked; a teaching ranking.)

| Drug | Sedation | Weight | EPS | Anti-Ch | ↓ BP | Prolactin | QTc |
|---|---|---|---|---|---|---|---|
| **Haloperidol** | ● | ● | ●●● | — | ● | ●●● | ●● |
| **Chlorpromazine** | ●●● | ●● | ●● | ●● | ●●● | ●●● | ● |
| **Clozapine** | ●●● | ●●● | — | ●●● | ●●● | — | ● |
| **Olanzapine** | ●● | ●●● | ● | ● | ● | ● | — |
| **Quetiapine** | ●●● | ●● | — | ● | ●● | — | ● |
| **Risperidone** | ● | ●● | ●● | — | ●● | ●●● | ● |
| **Ziprasidone** | ● | — | ● | — | ● | ● | ●●● |
| **Lurasidone** | ● | — | ●● | — | — | ● | — |
| **Aripiprazole** | — | ● | ●● | — | — | — | — |
| **Brexpiprazole** | ● | ●● | ● | — | ● | — | — |
| **Cariprazine** | — | — | ●● | — | — | — | — |

**Champions:**

- **EPS:** haloperidol
- **Sedation/weight/anticholinergic/↓ BP:** clozapine (and chlorpromazine)
- **Weight:** olanzapine, clozapine
- **Prolactin:** haloperidol, chlorpromazine, risperidone
- **QTc:** ziprasidone

Partial-agonist EPS is mainly **akathisia** (most with **cariprazine**, least with **brexpiprazole**).




*Slide 39: Antipsychotic side-effect map*

## 2.16 EPS: Timeline and Acute Dystonia (Slides 40–43) ⭐⭐

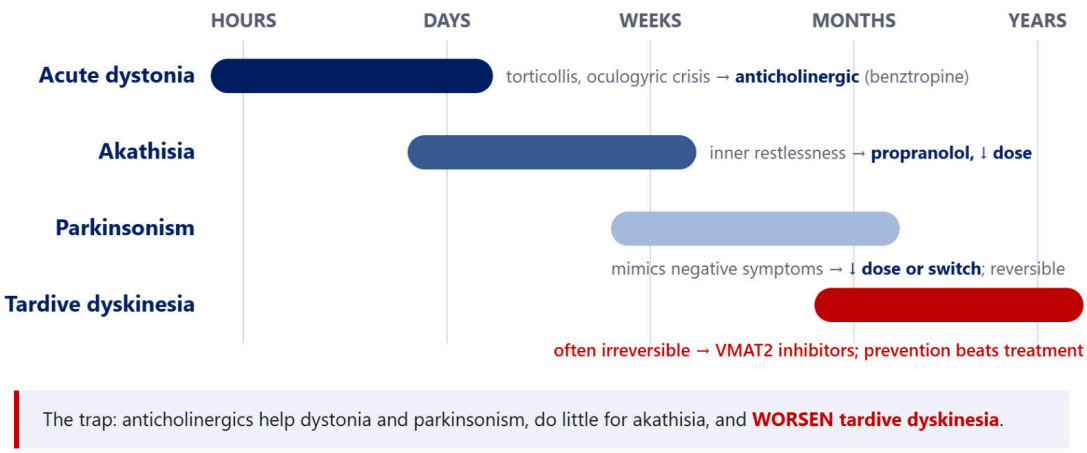
**Case (slides 40–41):**

- 19-year-old given **haloperidol** in the ED
- **hours later: neck twists to one side**, **eyes roll up and lock**; frightened but **fully conscious**

→ ⭐ **B: acute dystonia → anticholinergic (benztropine or diphenhydramine).** Young men are at highest risk.

| Syndrome | Onset | Features | Treatment |
|---|---|---|---|
| ⭐ **Acute dystonia** | **Hours–days** | **Torticollis, oculogyric crisis**, truncal spasm | ⭐ **IM/IV anticholinergic (benztropine)**: reverses in minutes |
| ⭐ **Akathisia** | **Days–weeks** | Inner restlessness | ⭐ **Propranolol, ↓ dose** |
| **Parkinsonism** | **Weeks–months** | Mimics negative symptoms | **↓ Dose or switch**; reversible |
| ⭐ **Tardive dyskinesia** | **Months–years** | Choreoathetoid, orofacial | ⭐ **Often irreversible → VMAT2 inhibitors**; prevention beats treatment |

- ⭐ **THE TRAP:** anticholinergics **help dystonia and parkinsonism**, **do little for akathisia**, and ⭐ **WORSEN tardive dyskinesia**.
- **Highest TD risk** with **long-term high-potency FGAs**.
- **Preserved consciousness** separates acute dystonia from a **seizure**.




*The trap: anticholinergics help dystonia and parkinsonism, do little for akathisia, and WORSEN tardive dyskinesia. Highest TD risk with long-term high-potency FGAs.*

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*Slide 42: EPS timeline*

**The two signs from the vignette, made visible: the eyes locked upward and the neck and trunk thrown into spasm — hours after a high-potency antipsychotic, and fully conscious throughout.**

THE EYES



**Oculogyric crisis**

Sustained conjugate UPWARD deviation of both eyes

THE NECK & TRUNK



**Acute dystonic reaction**

Torticollis & truncal spasm · dystonic arm posturing

*Preserved consciousness separates acute dystonia from a seizure; it reverses in minutes with an IM/IV anticholinergic. Eyes: Tokioka K et al., Clin Case Rep 2026, CC BY 4.0. Posture: James Heilman MD, Wikimedia Commons, CC BY-SA 3.0.*

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*Slide 43: Acute dystonia (oculogyric crisis, torticollis)*

## 2.17 Metabolic, Prolactin, QT and Other Effects (Slides 45–47)

**Metabolic:** driven by **H1, 5-HT2C, D2 antagonism** + leptin changes.

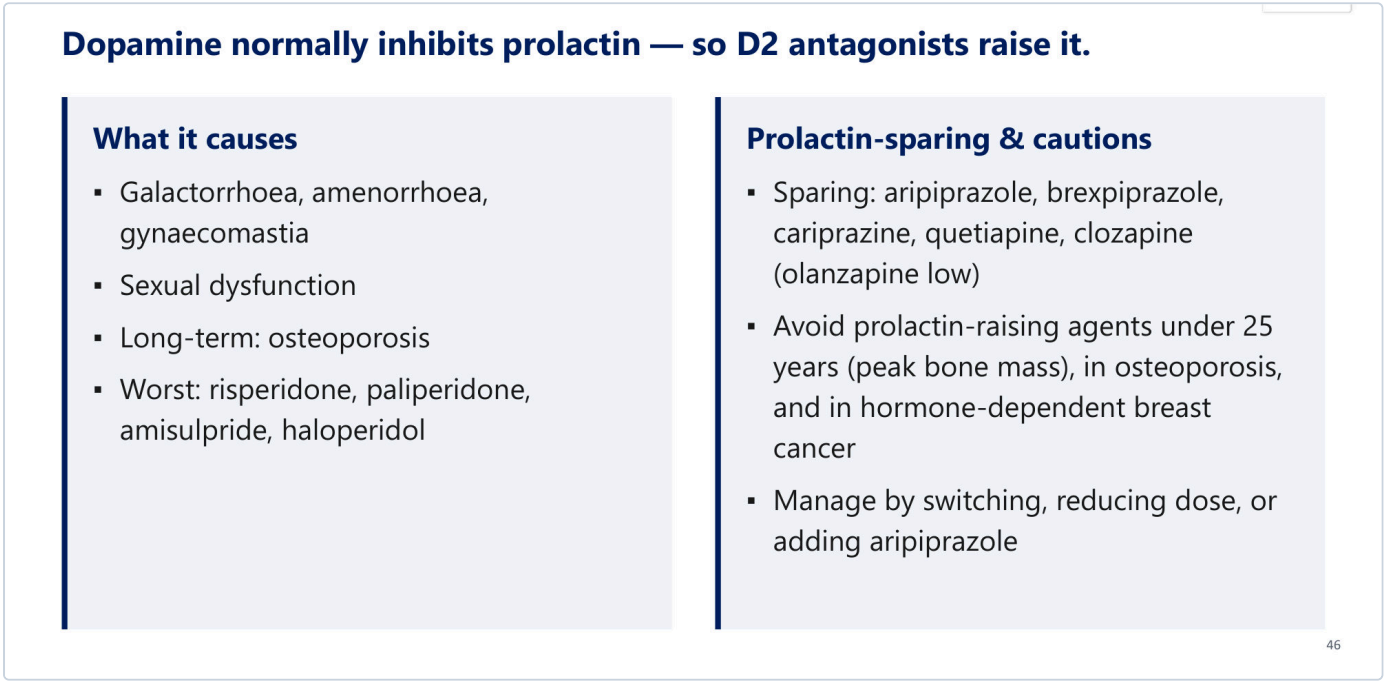
| Risk | Agents | Action |
|---|---|---|
| ⭐ **High** | **Clozapine, olanzapine** | Expect significant gain; monitor from baseline |
| **Moderate** | Quetiapine, risperidone, brexpiprazole | Monitor; consider switching |
| ⭐ **Low/neutral** | **Aripiprazole, cariprazine, ziprasidone, lurasidone** | Preferred where metabolic risk exists |

⭐ **Monitoring (all antipsychotics):**

1. baseline
2. **weight weekly × 6 weeks**
3. weight, waist, **glucose/HbA1c, lipids, BP at 12 weeks**
4. then **1 year and annually**

**Management:** switch to a weight-neutral agent; ⭐ **consider metformin**; lifestyle; cardiometabolic referral.

| Hyperprolactinaemia | QT and other |
|---|---|
| **Galactorrhoea, amenorrhoea, gynaecomastia, sexual dysfunction**; long-term ⭐ **osteoporosis** | Antipsychotics block cardiac **$K^+$ channels**; ⭐ **> 500 ms → torsades risk climbs** |
| ⭐ **Worst: risperidone, paliperidone, amisulpride, haloperidol** | ECG + electrolytes in at-risk patients |
| ⭐ **Sparing: aripiprazole, brexpiprazole, cariprazine, quetiapine, clozapine** (olanzapine low) | ⭐ **Higher risk: haloperidol (esp. IV), ziprasidone, pimozide** |
| ⭐ **Avoid prolactin-raisers under 25** (peak bone mass), in osteoporosis, and in **hormone-dependent breast cancer** | Anticholinergic effects; **seizures (chlorpromazine, clozapine)**; orthostatic hypotension; **SIADH** |
| Manage: switch, reduce, or **add aripiprazole** | — |




*Slide 46: Hyperprolactinaemia*

## 2.18 Clozapine (Slides 48–50) ⭐⭐

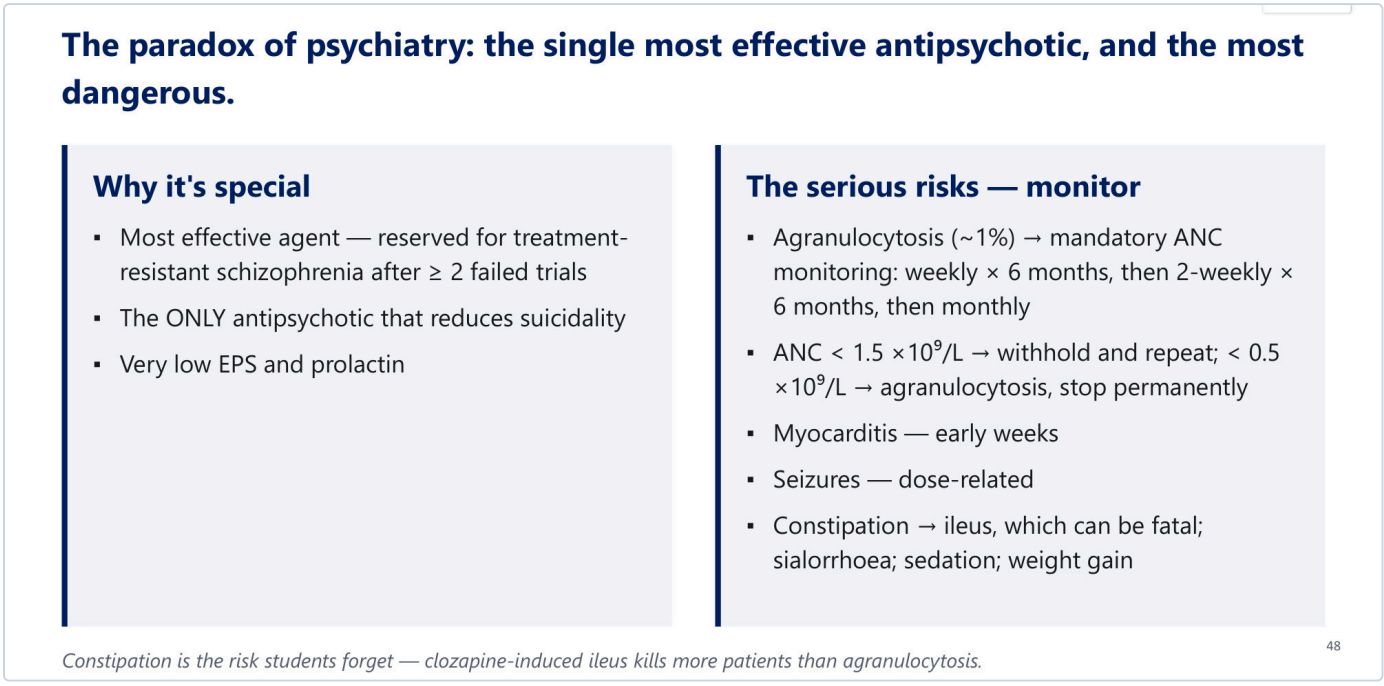
| Why it's special | The serious risks |
|---|---|
| ⭐ **Most effective**: treatment-resistant schizophrenia after **≥ 2 failed trials** | ⭐ **Agranulocytosis (~1%)** → mandatory ANC: **weekly × 6 months, 2-weekly × 6 months, then monthly** |
| ⭐ **The ONLY antipsychotic that reduces suicidality** | ⭐ **ANC < 1.5 × $10^9$/L → withhold and repeat; < 0.5 → agranulocytosis, stop permanently** |
| **Very low EPS and prolactin** | **Myocarditis** (early weeks) |
| — | **Seizures** (dose-related) |
| — | ⭐ **Constipation → ileus (can be fatal)**; **sialorrhoea**; sedation; weight gain |

⭐ **Clozapine-induced ileus kills more patients than agranulocytosis.**

**Case (slides 49–50):**

- **4 months** on clozapine
- **T 38.6 °C + sore throat** since yesterday; looks well; chest clear

→ ⭐ **B: urgent FBC and ANC TODAY.** Withhold clozapine until the result is back. "Reassure, wait a week" can be fatal.




*Slide 48: Clozapine*

## 2.19 LAIs and Special Populations (Slide 51)

| **Long-acting injectables** | ⭐ **Dementia: black box** | **Renal & liver** |
|---|---|---|
| FGA: **haloperidol, fluphenazine decanoate** | ⭐ **Increased mortality in elderly patients with dementia** (CV events, falls) | **Renal:** avoid **sulpiride/amisulpride** and strong anticholinergics; consider haloperidol or low-dose olanzapine |
| SGA: **risperidone, paliperidone, aripiprazole, olanzapine** | Slide figures "≈ 2.6% vs ≈ 4.5%" over 10-week trials (*[added]* = **placebo 2.6% vs drug 4.5%**; the slide's order is ambiguous) | **Liver:** monitor LFTs; transaminitis in the first **1–6 weeks** |
| ⭐ **Transform adherence** in relapsing illness | Use only if essential, lowest dose | — |

🔁 **Recap: Antipsychotics.**

- **NMS:** days; rigidity, fever, ↑ CK → stop, dantrolene/bromocriptine; hold ≥ 5 days.
- **FGA potency:** high = EPS/prolactin; low = anticholinergic/sedation/↓ BP.
- **SGA:** 5-HT2A + D2, metabolic. **Partial agonists:** aripiprazole, brexpiprazole, cariprazine (akathisia).
- **Map champions:** haloperidol EPS; clozapine/olanzapine weight; risperidone prolactin; ziprasidone QT.
- **EPS clock:** dystonia (anticholinergic) → akathisia (propranolol) → parkinsonism → TD (VMAT2; anticholinergics worsen it).
- **Metabolic monitoring:** baseline, weekly × 6, 12 weeks, annually; metformin.
- **Clozapine:** ANC (< 1.5 hold, < 0.5 stop); fever/sore throat → ANC today; ileus; only anti-suicide antipsychotic.
- **Dementia black box.**

▶️ **Up next: Mood stabilisers.**

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# PART 4: MOOD STABILISERS

## 2.20 Lithium (Slides 53–56) ⭐⭐

**Case (slides 53–54):**

- 45-year-old on long-term lithium starts **ibuprofen** during a **hot week**
- **coarse tremor, unsteady gait, vomiting, slurred speech, confusion**

→ ⭐ **A: lithium toxicity.** NSAIDs, dehydration, thiazides and ACE inhibitors ↓ lithium clearance. **Hold lithium, hydrate, check the level; dialyse if severe.**

| Dosing & levels | Side-effects |
|---|---|
| ~**900–1800 mg/day**, divided | **Thyroid** abnormalities (*[added]* hypothyroidism, goitre); **benign leukocytosis** |
| ⭐ **Therapeutic 0.6–1.2 mEq/L; maintenance 0.6–0.8** | ⭐ **Polyuria/polydipsia: nephrogenic diabetes insipidus** |
| ⭐ **1.2–1.5 = ABOVE RANGE** (recheck, review dose); ⭐ **toxic > 1.5** | Weight gain; ⭐ **FINE tremor** (therapeutic) vs **COARSE** tremor (toxic); cognitive slowing |
| ⭐ **Monitor:** level, **renal function, TSH, Ca/PTH, ECG** | **Acne, hair loss** |
| **Gold standard; the only mood stabiliser that clearly reduces suicide** | ⭐ **Teratogenic: Ebstein's anomaly** (small risk) |

**Lithium toxicity by stage (slide 56):**

| Stage | Features |
|---|---|
| **Early** | Lethargy, ⭐ **COARSE tremor**, vomiting, diarrhoea |
| **Moderate** | **Confusion, dysarthria, nystagmus**, ECG changes |
| **Late** | **Seizures, hyperreflexia, renal failure, coma** → ⭐ **dialyse if severe** |

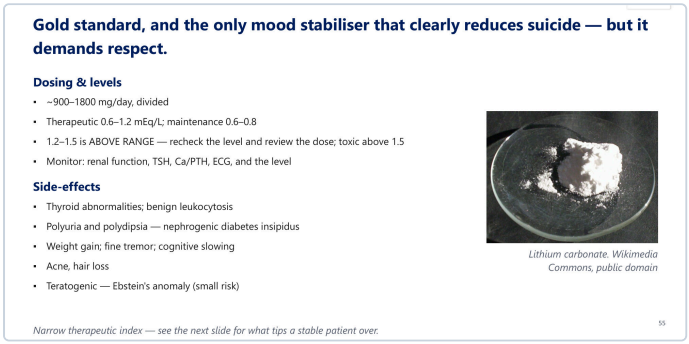
⭐ **What tips a stable patient over:**

- **dehydration**
- **NSAIDs**
- **thiazides**
- **ACE inhibitors**
- **renal impairment**

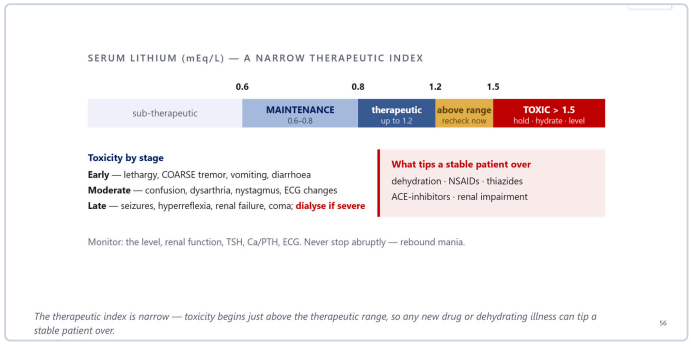
**Never stop abruptly** (rebound mania).

**Mnemonic *[added]* for lithium side-effects, "LITHIUM":**

- **L**eukocytosis
- **I**nsipidus (nephrogenic DI)
- **T**remor/**T**eratogen (Ebstein)
- **H**ypothyroidism/**H**yperparathyroidism
- **I**ncreased weight
- **U**rine (polyuria)
- **M**outh (metallic taste)




*Slide 55: Lithium dosing and monitoring*



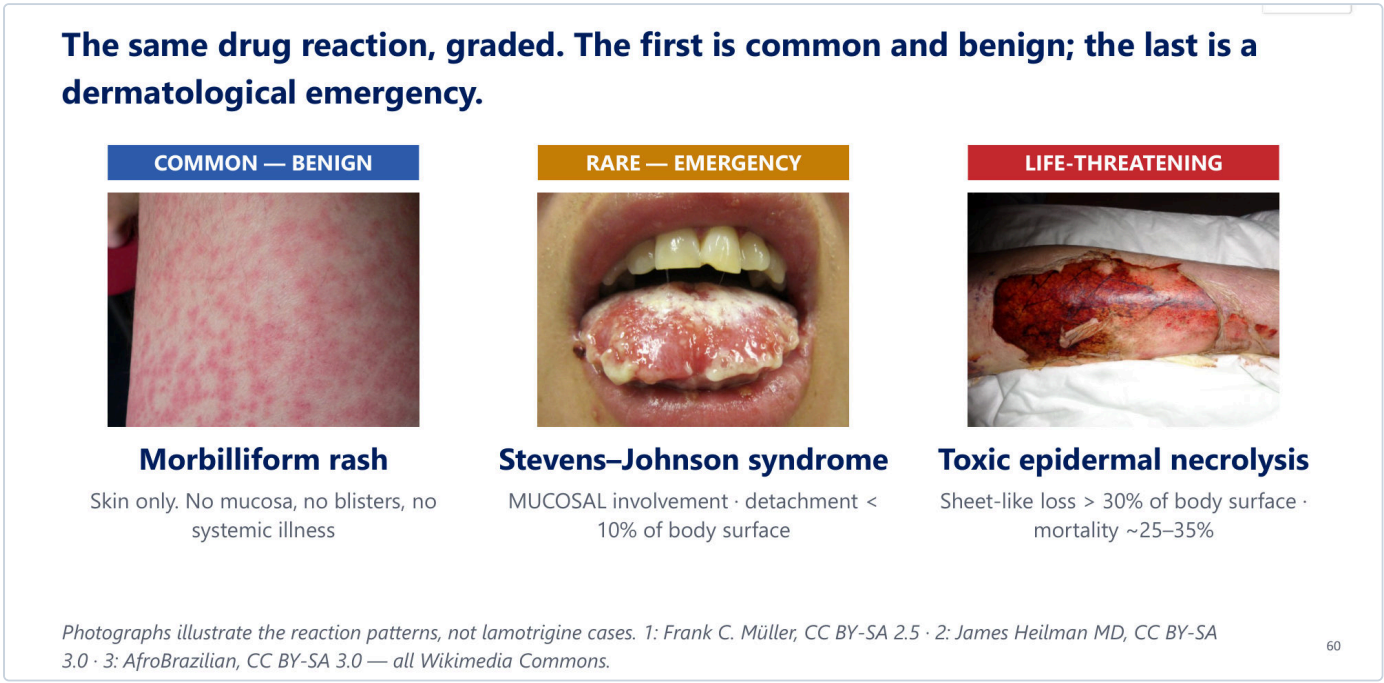

*Slide 56: Lithium toxicity and interactions*

## 2.21 Valproate, Carbamazepine, Lamotrigine (Slides 57–60) ⭐

| | Valproate | Carbamazepine | Lamotrigine |
|---|---|---|---|
| Best for | ⭐ **Acute mania; mixed and rapid-cycling states** | Mania (*[added]*), epilepsy, trigeminal neuralgia | ⭐ **Bipolar DEPRESSION and maintenance**; weak for acute mania |
| Key side-effects | **Thrombocytopenia**, platelet dysfunction; nausea, **weight gain**; ⭐ **hepatotoxicity**; sedation, tremor, **hair loss**; ⭐ **pancreatitis; hyperammonaemia** | ⭐ **Rash (most common)**; GI upset, transaminitis; sedation, dizziness, ataxia, **diplopia**; **AV conduction delay**; ⭐ **hyponatraemia (SIADH-like)**; ⭐ **aplastic anaemia/agranulocytosis** (rare) | Generally well tolerated; **no routine levels**; ⭐ **rash → SJS/TEN** |
| Pregnancy | ⭐ **Most teratogenic: neural-tube defects 1–2%** vs 0.14–0.2% background (folate-related); ↓ **childhood IQ**. **Avoid in women of childbearing potential** | *[added]* Also teratogenic (NTD) | — |
| ⭐ Pearl | — | ⭐ **HLA-B*1502 → SJS/TEN** in **Han Chinese and South-East Asian** patients: **GENOTYPE BEFORE STARTING**. ⭐ **Auto-induces CYP3A4** (its own level falls over weeks); **strong enzyme inducer** | ⭐ **VALPROATE DOUBLES LAMOTRIGINE LEVELS** (inhibits glucuronidation) → **slower titration, lower dose**. ⭐ **Any rash: stop and assess**; don't watch and wait |

**Lamotrigine rash severities (slide 60):**

| Rash | Features |
|---|---|
| **Morbilliform** (common, benign) | Skin only; no mucosa, blisters or systemic illness |
| ⭐ **SJS** (rare, emergency) | **Mucosal involvement**; detachment **< 10%** BSA |
| ⭐ **TEN** (life-threatening) | Sheet-like loss **> 30%** BSA; **mortality ~25–35%** |




*Slide 60: Lamotrigine rash, three severities*

🔁 **Recap: Mood stabilisers.**

- **Lithium:** 0.6–1.2 (maintenance 0.6–0.8); > 1.5 toxic. Coarse tremor → ataxia → confusion → seizures/coma (dialyse). NSAIDs, thiazides, ACEi, dehydration. Monitor renal, TSH, Ca/PTH, ECG. Nephrogenic DI; Ebstein; anti-suicide.
- **Valproate:** mania/mixed/rapid cycling; liver, pancreas, platelets, $NH_3$; **NTDs, ↓ IQ**.
- **Carbamazepine: HLA-B*1502**; auto-induction; SIADH; aplastic anaemia.
- **Lamotrigine:** bipolar depression; SJS; slow titration; **valproate doubles the level**.

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# PART 5: INTEGRATION

## 2.22 Cytochrome P450 Interactions (Slide 62) ⭐

| Inhibitors: raise other drugs | Inducers: lower other drugs |
|---|---|
| ⭐ **Fluoxetine, paroxetine → CYP2D6:** raise TCAs, some antipsychotics; ⭐ **codeine becomes ineffective** | ⭐ **Carbamazepine:** potent inducer; lowers the **OCP**, **lamotrigine**, many drugs; auto-induces |
| ⭐ **Fluvoxamine → CYP1A2:** raises **clozapine** and **theophylline** | ⭐ **Smoking induces CYP1A2** → lowers **clozapine and olanzapine** |
| Serotonergic combinations → serotonin syndrome | ⭐ **If the patient QUITS smoking, those levels RISE** (a classic trap) |

⭐ **Fewest CYP interactions: sertraline, escitalopram, mirtazapine.** Use them in polypharmacy.

**Sertraline, escitalopram and mirtazapine have the fewest CYP interactions — useful in the medically complex or polypharmacy patient.**

### Inhibitors — raise other drug levels

- Fluoxetine, paroxetine → CYP2D6: raise TCAs and some antipsychotics; codeine becomes ineffective
- Fluvoxamine → CYP1A2: raises clozapine and theophylline
- Watch serotonergic combinations → serotonin syndrome

### Inducers — lower other drug levels

- Carbamazepine — potent inducer: lowers the oral contraceptive, lamotrigine and many drugs; auto-induces
- Smoking induces CYP1A2 → lowers clozapine and olanzapine
- If the patient QUITS smoking, those levels RISE — a classic trap

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*Slide 62: CYP450 interactions*

## 2.23 Psychotropics on a Non-Psychiatric Ward (Slide 63) ⭐⭐

**"If you remember one slide as a future surgeon or physician, remember this one."**

| Rule | Detail |
|---|---|
| ⭐ **Reconcile on admission** | Psychotropics are among the **most commonly omitted** and **most dangerous to omit**. Check what the patient actually takes |
| ⭐ **Never stop abruptly** | **Benzodiazepines → withdrawal seizures**; **antidepressants → discontinuation**; ⭐ **clozapine > ~48 h off → re-titrate from a low dose** |
| ⭐ **NBM and dehydrated** | The classic **lithium-toxicity** setting: surgery, vomiting, diarrhoea, hot weather, a new **NSAID/thiazide/ACEi**. **Hold lithium, rehydrate, check the level** |
| ⭐ **Fever on clozapine** | **FBC + ANC the same day**. And take **constipation** seriously: ileus kills more than agranulocytosis |

**You will rarely start these drugs. You will constantly look after patients already taking them — and that is where the harm happens.**

### Reconcile on admission

Psychotropics are among the most commonly omitted drugs on an admission chart, and the most dangerous to omit. Check what the patient actually takes, and write it up.

### Never stop abruptly

Benzodiazepines → withdrawal seizures. Antidepressants → discontinuation syndrome. Clozapine → more than ~48 hours off means re-titrating from a low dose.

### Nil by mouth & dehydrated

The classic lithium-toxicity setting: surgery, vomiting, diarrhoea, hot weather, or a new NSAID / thiazide / ACE-inhibitor. Hold the lithium, rehydrate, check the level.

### Fever on clozapine

FBC and ANC the same day — agranulocytosis until proven otherwise. And take constipation seriously: clozapine ileus kills more patients than agranulocytosis does.

*If you remember one slide from this lecture as a future surgeon or physician rather than a psychiatrist, remember this one.*

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*Slide 63: Psychotropics on a non-psychiatric ward*

## 2.24 The Interactions You Will Cause (Slide 64) ⭐⭐

| You prescribe | Hits | What happens | Do |
|---|---|---|---|
| ⭐ **Tramadol** | SSRI/SNRI | **Serotonin syndrome** (the commonest trigger in practice) | Choose a **non-serotonergic analgesic** |
| ⭐ **Codeine** | **Fluoxetine, paroxetine** | CYP2D6 inhibition → **no conversion to morphine → NO analgesia** | Use another analgesic |
| ⭐ **NSAIDs** | **Lithium / SSRI** | **Lithium toxicity**; with SSRIs, **GI bleeding** | Avoid; if unavoidable, check the level and add gastric protection |
| **Clarithromycin, linezolid** | Clozapine / serotonergics | Enzyme inhibition **↑ clozapine**; ⭐ **linezolid is an MAOI → serotonin syndrome** | Check interactions before antibiotics |
| **Any QT-prolonging drug** | Antipsychotics | **Additive QT → torsades** | ECG and electrolytes before and after |

**Also: metoclopramide and ondansetron are serotonergic**; ⭐ **metoclopramide is a dopamine antagonist → EPS and can contribute to NMS**.

**These are not exotic psychiatric drugs — they are the everyday drugs YOU will prescribe, colliding with the psychotropic the patient is already on.**

| What you prescribe | Hits | What happens | What to do |
|---|---|---|---|
| **Tramadol** | SSRI / SNRI | Serotonin syndrome — the commonest trigger in practice | Choose a non-serotonergic analgesic |
| **Codeine** | Fluoxetine, paroxetine | CYP2D6 inhibition blocks conversion to morphine — NO analgesia | Use a different analgesic; codeine will simply not work |
| **NSAIDs** | Lithium / SSRI | Lithium level rises → toxicity. With SSRIs, GI bleeding risk rises | Avoid; if unavoidable, check the level and add gastric protection |
| **Clarithromycin, linezolid** | Clozapine / serotonergics | Enzyme inhibition raises clozapine; linezolid is an MAOI → serotonin syndrome | Check interactions before prescribing an antibiotic |
| **Any QT-prolonging drug** | Antipsychotics | Additive QT prolongation → torsades | ECG and electrolytes before and after |

*Also: metoclopramide and ondansetron are serotonergic, and metoclopramide is a dopamine antagonist that causes EPS and can contribute to NMS.*

64

*Slide 64: The interactions you will cause*

## 2.25 Black-Box Warnings (Slide 65) ⭐

| Drug/class | Black box |
|---|---|
| ⭐ **All antidepressants** | **↑ Suicidal ideation/behaviour in patients < 25, early in treatment** |
| ⭐ **All antipsychotics** | **↑ Mortality in elderly patients with dementia-related psychosis** |
| ⭐ **Clozapine** | **Agranulocytosis · myocarditis · seizures · orthostatic hypotension/syncope** |
| ⭐ **Valproate** | **Hepatotoxicity · pancreatitis · teratogenicity (NTD)** |
| ⭐ **Carbamazepine** | **Aplastic anaemia/agranulocytosis · SJS/TEN (HLA-B*1502)** |

These shape **prescribing and consent**: they are what you must tell the patient.

| Drug / class | Black-box warning |
|---|---|
| **All antidepressants** | Increased suicidal ideation and behaviour in patients under 25, early in treatment |
| **Antipsychotics (all)** | Increased mortality in elderly patients with dementia-related psychosis |
| **Clozapine** | Agranulocytosis · myocarditis · seizures · orthostatic hypotension and syncope |
| **Valproate** | Hepatotoxicity · pancreatitis · teratogenicity (neural-tube defects) |
| **Carbamazepine** | Aplastic anaemia / agranulocytosis · SJS/TEN (HLA-B*1502) |

*These shape both prescribing and consent — they are what you must tell the patient.*

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*Slide 65: Black-box warnings*

## 2.26 Integrative Case 1: Mr. H, 34 (Slides 66–67) ⭐

**Presentation:**

- **paroxetine 8 months** + **tramadol 2 days ago** + OTC **dextromethorphan** cough syrup
- **hours:** agitated, drenched, **HR 124, BP 158/92, 38.4 °C**
- **dilated pupils, hyperactive bowel sounds**, tremor, **ankle clonus (legs > arms)**, brisk reflexes
- **no rigidity**, no antipsychotic

| Question | Model answer (lecture) |
|---|---|
| 1. Diagnosis, culprits? | ⭐ **Serotonin syndrome**: paroxetine + **tramadol + dextromethorphan** (the last two most often overlooked) |
| 2. vs NMS? | **Hours vs days**; **clonus/hyperreflexia (legs)** vs lead-pipe rigidity; **dilated pupils**; **hyperactive vs reduced bowel sounds**; serotonergic vs antipsychotic |
| 3. Immediate management? | **Stop every serotonergic drug**; IV fluids, **active cooling**, cardiac monitoring, **benzodiazepines** |
| 4. Antidotes? | ⭐ **Cyproheptadine** for SS; **dantrolene and bromocriptine** for NMS: **don't swap them** |
| 5. Discharge? | Document as a drug interaction; warn that **tramadol and dextromethorphan are serotonergic**; plan the **washout** before any new serotonergic agent |

## 2.27 Integrative Case 2: Ms. K, 27 (Slides 68–69) ⭐

**Presentation:**

- **clozapine 400 mg for 2 years** (two antipsychotics failed before)
- **no bowel opening for 6 days**; distended, vomiting for 2 days
- **procyclidine started 3 weeks ago** for tremor
- **tympanic abdomen, almost no bowel sounds**; T 37.4 °C, HR 104
- ANC normal; not rigid; oriented
- the surgical registrar asks to "simply stop clozapine"

| Question | Model answer (lecture) |
|---|---|
| 1. Diagnosis? | ⭐ **Clozapine-induced GI hypomotility → ileus**, aggravated by **anticholinergic procyclidine** |
| 2. Why dangerous? | ⭐ **Commoner than agranulocytosis and higher mortality**, but **no mandatory monitoring**, so it is missed |
| 3. Management? | **Surgical assessment + imaging**, **stop the anticholinergic**, aggressive laxatives, **NBM if obstructed**; treat as a **surgical emergency** |
| 4. Stop clozapine? | **Not casually.** Stopping > 48 h → **re-titrate from a low dose**; she has failed two others. **Decide jointly with psychiatry** |
| 5. Prevention? | **Ask about bowels at every clozapine review**; **prophylactic laxatives**, not an added anticholinergic |

⭐ **Transferable lesson:** "the risk that is counted gets watched; the risk that is not counted gets missed."

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# 3. High-Yield Comparison Tables

## Table A: Antidepressant Selection Cheat-Sheet

| Clinical need | Choose | Avoid |
|---|---|---|
| Suicidal / overdose risk | SSRI (sertraline, fluoxetine) | **TCAs** (lethal) |
| Cardiac disease / QT | **Sertraline**, fluoxetine, mirtazapine | Citalopram/escitalopram (QT), TCAs |
| Insomnia + poor appetite | **Mirtazapine** | Bupropion (activating) |
| Sexual dysfunction concern | **Bupropion**, mirtazapine | SSRIs/SNRIs |
| Smoking cessation | **Bupropion** | — |
| Eating disorder / head injury / epilepsy | SSRI | **Bupropion** (seizures) |
| Neuropathic pain | **Duloxetine** (or TCA, e.g. amitriptyline *[added]*) | — |
| Polypharmacy | **Sertraline, escitalopram, mirtazapine** | Fluoxetine/paroxetine (2D6), fluvoxamine (1A2) |
| Elderly | Sertraline/mirtazapine; check $Na^+$ | TCAs (anticholinergic, falls), paroxetine |
| Breastfeeding | **Sertraline**, paroxetine | — |
| Pregnancy, first trimester | Sertraline/fluoxetine (risk–benefit) | Paroxetine (historic signal) |
| Atypical / resistant depression | MAOI (with tyramine diet) | — |
| Postpartum depression | **Zuranolone** | — |
| TRD / acute suicidality | **Esketamine** | — |
| Hypnotic | **Trazodone** (warn: priapism) | — |

## Table B: The Toxic Syndromes

| | Serotonin syndrome | NMS | Lithium toxicity | TCA overdose | Anticholinergic *[added]* |
|---|---|---|---|---|---|
| Onset | **Hours** | **Days–weeks** | Days (precipitant) | Hours | Hours |

| | Serotonin syndrome | NMS | Lithium toxicity | TCA overdose | Anticholinergic *[added]* |
|---|---|---|---|---|---|
| Neuro | **Clonus, hyperreflexia** | **Lead-pipe rigidity**, bradyreflexia | **Coarse tremor, ataxia**, dysarthria → seizures | Coma, seizures | Delirium |
| Pupils | **Dilated** | Normal | — | Dilated | Dilated |
| Bowel | **Hyperactive** | Reduced | Diarrhoea | Reduced | **Absent** |
| Skin | Sweaty | Sweaty | — | Dry | **Dry, hot** |
| Key lab | — | **↑ CK** | **Li > 1.5** | **QRS > 100 ms, R in aVR** | — |
| Rx | **Stop, benzodiazepines, cyproheptadine** | **Stop, dantrolene/bromocriptine** | **Hold, fluids, dialysis** | **IV bicarbonate** | Supportive (physostigmine) |

## Table C: Benzodiazepine Facts

| Item | Detail |
|---|---|
| Mechanism | GABA-A, **↑ frequency** (barbiturates: ↑ duration) |
| Liver-safe | **LOT**: lorazepam, oxazepam, temazepam |
| Long-acting | Diazepam, chlordiazepoxide, clonazepam |
| Tolerance | ~2 weeks |
| Max duration | 2–4 weeks |
| Withdrawal | Seizures, coma, death → **taper** |
| Reversal | Flumazenil (seizure risk; rarely used) |
| Pregnancy | Cleft palate signal; floppy infant; neonatal withdrawal |
| Non-dependence alternative | **Buspirone** (5-HT1A; 1–2-week onset) |

## Table D: Mood Stabilisers

| | Lithium | Valproate | Carbamazepine | Lamotrigine |
|---|---|---|---|---|
| Best for | Maintenance, **anti-suicide** | **Acute mania, mixed, rapid cycling** | Mania, epilepsy | **Bipolar depression**, maintenance |
| Level | **0.6–1.2** (toxic > 1.5) | — | — | Not routine |
| Signature risk | Nephrogenic DI, hypothyroid, ↑ Ca, **toxicity** | Liver, pancreas, platelets, $NH_3$ | **HLA-B*1502 SJS**, SIADH, aplastic anaemia | **SJS/TEN** |
| Pregnancy | **Ebstein's** | **NTD 1–2%, ↓ IQ** (worst) | NTD *[added]* | Relatively safer *[added]* |
| Interaction | NSAIDs, thiazides, ACEi ↑ level | **Doubles lamotrigine** | **Inducer** (OCP fails), auto-induction | ↑ by valproate |

### Table E: Antipsychotic Choice by Patient

| Patient factor | Prefer | Avoid |
|---|---|---|
| Metabolic risk/obesity/diabetes | **Aripiprazole, cariprazine, ziprasidone, lurasidone** | Olanzapine, clozapine |
| Prolactin/young woman/< 25/osteoporosis | **Aripiprazole**, quetiapine | Risperidone, paliperidone, amisulpride, haloperidol |
| Long QT/cardiac disease | Aripiprazole, lurasidone | **Haloperidol IV, ziprasidone, pimozide** |
| EPS-prone/Parkinson's *[added]* | **Quetiapine, clozapine** | Haloperidol |
| Bipolar depression | **Lurasidone, cariprazine** (quetiapine) | — |
| Treatment-resistant / suicidal | **Clozapine** | — |
| Non-adherent | **LAI** | — |
| Renal impairment | Haloperidol, low-dose olanzapine | Sulpiride, amisulpride |
| Elderly with dementia | Avoid if possible (black box) | — |

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## 4. Rules, Exceptions, Patterns

- **Antidepressants: equal efficacy**; choose on **side-effects + overdose safety**. No TCAs for the suicidal.
- **SSRI risks:** sexual dysfunction (ask), bleeding (NSAIDs), **SIADH (elderly; stop if Na < 125)**, citalopram QT (40 / 20 mg).
- **Short half-life (paroxetine, venlafaxine) → worst discontinuation**; fluoxetine needs no taper.
- **TCA overdose → IV bicarbonate** (QRS > 100 ms, R in aVR). Never flumazenil.
- **MAOI washout 2 weeks (5 after fluoxetine)**; tyramine crisis.
- **Bupropion lowers the seizure threshold** (eating disorders, head injury); **trazodone → priapism**.
- **Serotonin syndrome = fast + clonus; NMS = slow + rigidity.** Cyproheptadine vs dantrolene/bromocriptine; don't swap.
- **Hidden serotonergics:** tramadol, dextromethorphan, linezolid, triptans, ondansetron, metoclopramide, St John's wort.
- **Discontinuation (days, physical) vs relapse (weeks, mood).**
- **Continue 6–9 months after remission**: most relapses happen in continuation.
- **Benzodiazepines:** frequency; LOT; 2–4 weeks; taper; flumazenil → seizures; **reconcile on admission**.
- **Buspirone:** no dependence; slow.
- **FGA high-potency → EPS/prolactin; low-potency → anticholinergic/sedation/↓ BP.**
- **Anticholinergics: help dystonia/parkinsonism, not akathisia, WORSEN TD.**
- **Metabolic monitoring** schedule; metformin.
- **Clozapine:** ANC thresholds; fever → ANC today; **ileus > agranulocytosis**; **> 48 h gap → re-titrate**; smoking cessation ↑ level.
- **Lithium 0.6–1.2; > 1.5 toxic**; NSAIDs, thiazides, ACEi, dehydration; **fine tremor = therapeutic, coarse = toxic**.
- **Valproate: most teratogenic**; doubles lamotrigine.
- **Carbamazepine: HLA-B*1502**; inducer (OCP failure); auto-induction.
- **Lamotrigine: any rash → stop**; slow titration.

- **Black boxes:** antidepressants < 25; antipsychotics + dementia; clozapine; valproate; carbamazepine.

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## 5. Clinical Correlations

- **Surgery:**
  - reconcile psychotropics (benzodiazepine omission → seizures)
  - NBM/dehydration → **lithium toxicity**
  - clozapine gaps > 48 h → re-titrate
  - ileus on clozapine
- **Pain management:**
  - **tramadol + SSRI → serotonin syndrome**
  - **codeine useless with fluoxetine/paroxetine**
  - **NSAIDs + lithium → toxicity**
  - NSAIDs + SSRI → GI bleed
- **Infectious disease:**
  - **linezolid is an MAOI**
  - **clarithromycin raises clozapine**
  - fever on clozapine → ANC
- **Cardiology/ECG:**
  - citalopram, haloperidol IV, ziprasidone, pimozide → **QT**
  - TCA → **QRS widening, tall R in aVR** → bicarbonate
  - **clozapine myocarditis**
  - lithium ECG changes
- **Endocrinology:**
  - **SSRIs/carbamazepine → SIADH**
  - **lithium → nephrogenic DI, hypothyroidism, hyperparathyroidism/↑ Ca**
  - **antipsychotics → hyperprolactinaemia** (galactorrhoea, amenorrhoea, osteoporosis)
  - **SGAs → diabetes, dyslipidaemia** (metformin)
- **Gastroenterology: metoclopramide → EPS/NMS**; ondansetron is serotonergic; valproate → hepatotoxicity, pancreatitis.
- **Obstetrics:**
  - **valproate (NTD, ↓ IQ)** and **lithium (Ebstein's)**
  - benzodiazepines (floppy infant)
  - paroxetine in the first trimester
  - **zuranolone** for PPD
  - **sertraline** in breastfeeding
- **Dermatology: lamotrigine and carbamazepine SJS/TEN**; HLA-B*1502 screening.
- **Haematology:** clozapine agranulocytosis; carbamazepine aplastic anaemia; valproate thrombocytopenia; lithium leukocytosis.
- **Smoking cessation: bupropion, varenicline**. Quitting smoking **raises clozapine/olanzapine levels**.

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## 6. Rapid Revision (Last-Day)

1. **Antidepressants:** equal efficacy; **60–70% respond at 4–6 weeks**; choose by side-effects and overdose safety.
2. **SSRIs:**
   - **fluoxetine** (long t½), **sertraline** (cardiac-safe, few interactions)
   - **citalopram/escitalopram** (QT; citalopram max 40 mg, 20 mg if > 60)
   - **paroxetine** (worst discontinuation), **fluvoxamine** (CYP1A2 ↑ clozapine)
3. **SSRI harms:** sexual dysfunction, GI, anxiety; **bleeding** (+ NSAIDs), **SIADH** (stop if Na < 125), serotonin syndrome, discontinuation.
4. **SNRIs:** venlafaxine (↑ BP, discontinuation); duloxetine (pain; avoid in severe liver disease).
5. **TCAs:** H1 (sedation, weight), M (anticholinergic), α1 (orthostasis), **$Na^+$ (QRS, arrhythmia)**. Tertiary (amitriptyline, imipramine, clomipramine, doxepin) vs secondary (nortriptyline, desipramine). **Overdose → IV bicarbonate.**
6. **MAOIs:** tyramine crisis; serotonin syndrome; **washout 2 weeks / 5 weeks after fluoxetine**; moclobemide reversible.
7. **Bupropion** (NDRI: no sexual dysfunction/weight gain, smoking; **seizures**) · **mirtazapine** (NaSSA: sleep, appetite, no QT) · **trazodone** (SARI: hypnotic, **priapism**).
8. **New agents:** vortioxetine, vilazodone, agomelatine; **esketamine** (NMDA; TRD, suicidality; monotherapy 2025); **Auvelity** (2022); **zuranolone** (PPD, 2023).
9. **Serotonin syndrome:** hours; mental-status change, **clonus/hyperreflexia (legs)**, autonomic; tramadol, DXM, linezolid, triptans; **stop, cool, benzodiazepines, cyproheptadine**.
10. **NMS:** days; **lead-pipe rigidity**, fever, ↑ CK, myoglobinuria; high-potency FGAs, young men, dehydration; **stop, cool, dantrolene + bromocriptine**; hold ≥ 5 days.
11. **Discontinuation (FINISH):** days; brain zaps; < 2 weeks; restart and taper.
12. **Duration:** acute 6–12 weeks; **continuation 6–9 months after remission**; maintenance ≥ 2 years. Benzodiazepines 2–4 weeks; antipsychotics 1–2 years after a first episode; lithium long-term.
13. **Pregnancy/lactation:** SSRIs not major teratogens; avoid paroxetine in the first trimester; **sertraline/paroxetine lowest in milk**.
14. **Benzodiazepines:** - GABA-A **frequency** (barbiturates: duration) - **LOT** for liver disease/elderly - tolerance in 2 weeks; withdrawal **seizures/death** → taper - **flumazenil → seizures** - floppy infant
15. **Z-drugs** still cause dependence; **buspirone** = 5-HT1A, no dependence, 1–2-week onset.
16. **Pathways:** mesolimbic (therapeutic), mesocortical (worsens negative), nigrostriatal (EPS), TIDA (prolactin).
17. **FGA potency:** high (haloperidol, fluphenazine) = EPS/prolactin; low (chlorpromazine) = anticholinergic/sedation/↓ BP.
18. **SGA = 5-HT2A + D2**; **partial agonists:** aripiprazole, brexpiprazole, cariprazine. **Lurasidone/cariprazine** for bipolar depression. Efficacy is equal except **clozapine**; judge at ≥ 4 weeks.
19. **Map:** haloperidol EPS; clozapine/olanzapine weight; risperidone prolactin; ziprasidone QT; aripiprazole/cariprazine clean (akathisia).
20. **EPS:** - **dystonia** (hours, benztropine) - **akathisia** (days, propranolol/↓ dose) - **parkinsonism** (weeks, ↓ dose) - **TD** (months–years, VMAT2) - **anticholinergics worsen TD**
21. **Metabolic monitoring:** baseline, weekly × 6, 12 weeks, 1 year, annually; metformin.
22. **Prolactin:** worst risperidone/paliperidone/amisulpride/haloperidol; sparing aripiprazole etc.; avoid < 25.
23. **QT > 500 ms → torsades;** haloperidol IV, ziprasidone, pimozide.
24. **Clozapine:** - ≥ 2 failed trials; **only anti-suicide antipsychotic** - agranulocytosis ~1%; ANC weekly × 6 months → 2-weekly × 6 months → monthly; **< 1.5 hold; < 0.5 stop** - myocarditis; seizures; **ileus** (kills more); sialorrhoea - **fever/sore throat → ANC today**

25. **LAIs** for adherence; **dementia black box**.
26. **Lithium:** - **0.6–1.2 (maintenance 0.6–0.8)**; 1.2–1.5 above range; **> 1.5 toxic** - **coarse tremor → ataxia, dysarthria → seizures, coma → dialysis** - NSAIDs, thiazides, ACEi, dehydration - monitor level, renal, TSH, Ca/PTH, ECG - **nephrogenic DI**, hypothyroidism, leukocytosis, **Ebstein's**
27. **Valproate:** mania/mixed/rapid cycling; liver, pancreas, platelets, $NH_3$; **NTD 1–2%, ↓ IQ**.
28. **Carbamazepine:** rash, **SIADH**, AV block, aplastic anaemia; **HLA-B*1502**; **CYP3A4 auto-induction**; inducer.
29. **Lamotrigine:** bipolar depression; **SJS/TEN**; slow titration; **valproate doubles the level**.
30. **CYP:** fluoxetine/paroxetine 2D6 (↑ TCAs, codeine fails); fluvoxamine 1A2 (↑ clozapine); carbamazepine induces (OCP); **smoking induces 1A2; quitting ↑ clozapine**.
31. **Ward rules:** reconcile · never stop abruptly · NBM + lithium · fever on clozapine.
32. **Interactions you cause:** tramadol + SSRI · codeine + fluoxetine/paroxetine · NSAID + lithium/SSRI · linezolid (MAOI) · clarithromycin + clozapine · QT drugs · metoclopramide (EPS/NMS).
33. **Black boxes:** antidepressants (< 25 suicidality) · antipsychotics (dementia mortality) · clozapine · valproate · carbamazepine.

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## 7. Exam-Style Questions

**Q1 (lecture, slides 11–12).** A 22-year-old took a bottle of amitriptyline 2 h ago: drowsy, BP 88/54, HR 128; QRS 130 ms, tall R in aVR. Most important immediate treatment? A. Activated charcoal alone · **B. IV sodium bicarbonate** · C. IV flumazenil · D. IV naloxone

> **Answer: B** (lecture key).
> - $Na^+$-channel blockade; QRS > 100 ms + R in aVR predict seizures/arrhythmia.
> - Bicarbonate overcomes it. Flumazenil is wrong and dangerous.

**Q2 (lecture, slides 16–17).** A 30-year-old on sertraline is given tramadol; within hours he is agitated, confused, sweating, 39 °C, tachycardic, with brisk reflexes and clonus worse in the legs. Diagnosis? A. NMS · **B. Serotonin syndrome** · C. Anticholinergic toxicity · D. Malignant hyperthermia

> **Answer: B** (lecture key). Fast onset + mental-status change + autonomic instability + neuromuscular hyperactivity. Stop the drugs, cool, benzodiazepines, cyproheptadine if severe.

**Q3 (lecture, slides 29–30).** A 58-year-old on diazepam 20 mg/day for years (omitted from his admission chart) is tremulous, sweating, BP 175/100 on day 3 after hernia repair, then seizes. He doesn't drink. Diagnosis and need? A. Alcohol withdrawal → thiamine · **B. Benzodiazepine withdrawal → reinstate a benzodiazepine** · C. Post-op delirium → antipsychotic · D. Serotonin syndrome → cyproheptadine

> **Answer: B** (lecture key). Reinstate, then taper deliberately. Reconcile psychotropics on every admission.

**Q4 (lecture, slides 49–50).** A 29-year-old 4 months into clozapine has a fever of 38.6 °C and a sore throat since yesterday; looks well; chest clear. First action? A. Reassure: probably viral · **B. Urgent FBC and ANC today** · C. Antibiotics, review in a week · D. Stop clozapine and discharge

> **Answer: B** (lecture key).
> - Agranulocytosis until proven otherwise; withhold clozapine until the result is back.
> - ANC < 1.5 → withhold and repeat; < 0.5 → stop permanently.

**Q5 (lecture, slides 53–54).** A 45-year-old on long-term lithium starts ibuprofen during a hot week; she develops coarse tremor, unsteady gait, vomiting, slurred speech and confusion. What is happening? **A. Lithium toxicity** · B. Serotonin syndrome · C. Hypoglycaemia · D. Cerebellar stroke

> **Answer: A** (lecture key).
> - NSAIDs + dehydration ↓ clearance.
> - Hold lithium, hydrate, check the level, dialyse if severe.

**Additional lecture MCQs:**

- **Slides 21–22:** stopped paroxetine 3 days ago; dizziness, nausea, brain zaps, vivid dreams, mood unchanged → **discontinuation syndrome**. Restart and taper.
- **Slides 34–35:** haloperidol 5 days ago; fever, lead-pipe rigidity, fluctuating consciousness, labile BP, ↑ CK → **NMS**.
- **Slides 40–41:** haloperidol hours ago; torticollis and oculogyric crisis, fully conscious → **acute dystonia → benztropine**.

**Lecture board one-liners (slide 70):**

| # | Question | Answer |
|---|---|---|
| 1 | Fast onset, clonus, hyperreflexia after tramadol + SSRI? | **Serotonin syndrome** → benzodiazepines + cyproheptadine |
| 2 | Lead-pipe rigidity, fever, ↑ CK days after haloperidol? | **NMS** → stop, dantrolene/bromocriptine |
| 3 | Antidepressant to avoid in an eating disorder? | **Bupropion** (seizure threshold) |
| 4 | Benzodiazepines safe in liver disease? | **LOT**: lorazepam, oxazepam, temazepam |
| 5 | Genotype before carbamazepine in Han Chinese/SE Asian patients? | **HLA-B*1502** (SJS/TEN) |

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## 8. Exam Pitfalls & Traps

| Trap | Correct thinking |
|---|---|
| Choosing a TCA for a suicidal patient | ❌ **Lethal in overdose** |
| Flumazenil for an unknown overdose with wide QRS | ❌ Dangerous (seizures); TCA → **bicarbonate** |
| Citalopram 60 mg in a 70-year-old | ❌ Max **20 mg if > 60** (QT) |
| Ignoring sodium in an elderly patient on an SSRI | ❌ **SIADH**; stop if Na < 125 |
| SSRI + NSAID without gastric protection | ❌ ↑ GI bleeding |
| Tramadol for back pain in a patient on an SSRI | ❌ **Serotonin syndrome**; choose a non-serotonergic analgesic |
| Codeine for a patient on fluoxetine/paroxetine | ❌ CYP2D6 inhibition → **no analgesia** |
| Linezolid with an SSRI | ❌ Linezolid is an **MAOI** |
| Starting an SSRI the day after stopping an MAOI | ❌ **2-week washout** (5 weeks after fluoxetine) |

| Trap | Correct thinking |
|---|---|
| Bupropion for anxious depression / bulimia | ❌ Doesn't treat anxiety; **seizure risk** in eating disorders |
| Brain zaps 3 days after stopping paroxetine = relapse | ❌ **Discontinuation**: restart and taper |
| Stopping an antidepressant at recovery | ❌ Continue **6–9 months after remission** |
| Swapping cyproheptadine and dantrolene | ❌ Cyproheptadine = SS; dantrolene/bromocriptine = NMS |
| Rigidity + fever + clonus | ⚠️ Clonus/hyperreflexia → SS; lead-pipe/bradyreflexia → NMS |
| Benzodiazepines for months | ❌ **2–4 weeks**; tolerance by ~2 weeks |
| Omitting long-term diazepam on admission | ❌ Withdrawal **seizures**: reconcile |
| Diazepam in liver failure | ❌ Use **LOT** |
| Buspirone PRN for a panic attack | ❌ 1–2-week onset; useless acutely |
| Restless patient on an antipsychotic → more antipsychotic or benztropine | ❌ **Akathisia** → propranolol / ↓ dose (anticholinergics do little) |
| Benztropine for tardive dyskinesia | ❌ Anticholinergics **worsen TD**; use VMAT2 inhibitors |
| Torticollis + upward gaze = seizure | ❌ Conscious → **acute dystonia** → IM anticholinergic |
| Risperidone for a 20-year-old woman with amenorrhoea | ❌ Prolactin-raising; prefer **aripiprazole**; avoid < 25 |
| Olanzapine for an obese diabetic | ❌ Choose aripiprazole/lurasidone/ziprasidone/cariprazine |
| Fever on clozapine = viral | ❌ **ANC today** |
| Constipation on clozapine is trivial | ❌ **Ileus kills more than agranulocytosis**; prophylactic laxatives |
| Restarting clozapine at the full dose after 3 days off | ❌ **> 48 h → re-titrate** |
| Patient on clozapine quits smoking, no change needed | ❌ CYP1A2 de-induction → **levels rise** |
| Fine tremor on lithium = toxicity | ⚠️ Fine = therapeutic side-effect; **coarse** = toxicity |
| Lithium + NSAID/thiazide/ACEi without level checks | ❌ ↑ Lithium → toxicity |
| Valproate first-line in a young woman | ❌ **Most teratogenic** (NTD, ↓ IQ) |
| Carbamazepine in a Han Chinese patient without testing | ❌ **HLA-B*1502** first |
| Carbamazepine + combined OCP | ❌ Enzyme induction → **contraceptive failure** |
| Lamotrigine at the usual titration with valproate | ❌ Valproate **doubles** the level: slower, lower |
| A rash on lamotrigine: watch and wait | ❌ **Stop and assess** (SJS/TEN) |
| Metoclopramide is "just an antiemetic" | ❌ D2 antagonist (EPS, NMS) and serotonergic |
| Antipsychotic for agitation in dementia, freely | ❌ **Black box: ↑ mortality** |